noc22_cs107

Mock Exam 5 | Course ID: noc22_cs107

Q1. According to the lecture, what is the primary purpose of a Work Breakdown Structure (WBS)?

A) To calculate the project's total cost and budget.
B) To create a hierarchical decomposition of project work into smaller, manageable components.
C) To assign specific developers to each task in the project.
D) To determine the critical path of the project.

Source: AvwZ7JO8LNk | Difficulty: Easy

Q2. In the Scrum framework, what is a 'Sprint'?

A) A long-term project plan spanning several months.
B) A daily 15-minute meeting for the development team.
C) A fixed-duration iteration, typically one month long, to develop a product increment.
D) The final phase of the project where all features are integrated.

Source: 0JYkgfgKn8I | Difficulty: Easy

Q3. What does Parkinson's Law state in the context of project estimation?

A) The best way to estimate a project is to break it down into its smallest parts.
B) Work expands to fill the time available.
C) A project without a reliability requirement can meet any other requirement.
D) The more people you add to a late project, the later it gets.

Source: 0mpuhlabOWE | Difficulty: Easy

Q4. In project monitoring and control, what is a 'milestone'?

A) A daily task assigned to a junior developer.
B) A designated key event or significant point in a project's timeline.
C) The total budget allocated for a project phase.
D) An informal discussion about project progress.

Source: 3Rjj4IyO3hc | Difficulty: Easy

Q5. In Earned Value Analysis, what is 'Planned Value (PV)'?

A) The actual cost incurred for the work performed.
B) The total value credited to a project at a specific point in time.
C) The original budgeted cost for work scheduled to be completed by a given date.
D) The final profit earned at the end of the project.

Source: 88pXUUtmnbg | Difficulty: Easy

Q6. What is the Hawthorne Effect?

A) The tendency for project scope to expand over time.
B) The phenomenon where individuals improve performance in response to being observed.
C) The principle that 90% of execution time is spent in 10% of the code.
D) A management style that assumes employees are inherently lazy.

Source: ikOTZBPaklE | Difficulty: Easy

Q7. What is a 'functional organization' structure?

A) A structure where teams are formed for a single project and then disbanded.
B) A structure where employees are grouped by specialized functions like design or testing.

C) A hybrid structure with dual reporting lines to project and functional managers.
D) A flat structure where all decisions are made by consensus.
Source: MpNjo7bhBTg | Difficulty: Easy

Q8. What is a 'resource histogram' used for in project management?

A) To calculate the total cost of all project resources.
B) To list all activities on the critical path.
C) To visualize the demand for a specific resource over the project's timeline.
D) To document the project's communication plan.
Source: ZcO5UNzMtz8 | Difficulty: Easy

Q9. In contract management, what is a 'bespoke system'?

A) A generalized software package available for immediate purchase.
B) A software system specifically created and tailored for a single customer's unique requirements.
C) A core system that is purchased and then customized to meet specific needs.
D) A contract where the customer is charged per hour of work.
Source: olrW1uKQe0sM | Difficulty: Easy

Q10. What is meant by the 'intangibility' of software?

A) Software is very easy to change and modify.
B) Software does not consume physical space or have weight.
C) Software progress is difficult to visually assess until it is fully operational.
D) Software development is primarily intellectual work.
Source: ilHPCbkZLV4 | Difficulty: Easy

Q11. In the Incremental Model, how is the Value-to-Cost (V/C) ratio used?

A) To calculate the total net profit of the project.
B) To determine the number of developers needed for a task.
C) To prioritize which increments to develop next by comparing customer value to development cost.
D) To decide whether to use the Waterfall model or the RAD model.
Source: 0O6fJSkafOo | Difficulty: Medium

Q12. What is the primary role of the Scrum Master in a Scrum team?

A) To define features and prioritize the Product Backlog.
B) To write the code for the most complex user stories.
C) To act as a servant-leader who facilitates the process and removes impediments.
D) To represent the customer's interests and accept or reject completed work.
Source: 0JYkgfgKn8I | Difficulty: Medium

Q13. What is the key difference between Project Risk and Business Risk?

A) Project Risk relates to budget overruns, while Business Risk relates to schedule delays.
B) Project Risk threatens the successful execution of the project, while Business Risk threatens the benefits of the delivered project.
C) Project Risk is managed by the development team, while Business Risk is managed by the customer.
D) Project Risk is only relevant for technical projects, while Business Risk applies to all projects.
Source: 8RuJfk1Xj2U | Difficulty: Medium

Q14. How does Net Present Value (NPV) analysis improve upon evaluation techniques like ROI and Payback Period?

A) It is simpler to calculate and requires less data.
B) It provides a result as a percentage, which is easier to understand.
C) It ignores future cash flows, focusing only on the initial investment.
D) It accounts for the time value of money by discounting future cash flows.

Source: 8UlgpszPK6w | Difficulty: Medium

Q15. In a PERT/CPM network, what is the rule for calculating the Earliest Start (ES) time for an activity at a merge point (an activity with multiple predecessors)?

A) It is the average of the Earliest Finish (EF) times of all its predecessors.
B) It is the earliest (minimum) of the Earliest Finish (EF) times of all its predecessors.
C) It is the latest (maximum) of the Earliest Finish (EF) times of all its predecessors.
D) It is determined by the duration of the longest predecessor activity.

Source: 8w2gm0AO-KY | Difficulty: Medium

Q16. How does the V-Model emphasize verification and validation compared to the classical waterfall model?

A) It removes the testing phase entirely to speed up development.
B) It integrates test planning and design activities into every development phase.
C) It performs all testing only after the coding phase is 100% complete.
D) It relies on customer feedback instead of formal testing activities.

Source: J1-LpZr9GSQ | Difficulty: Medium

Q17. What is the purpose of 'refactoring' in agile development?

A) To add new features requested by the customer during an iteration.
B) To fix bugs discovered during the testing phase.
C) To restructure existing code without changing its external behavior, to improve its internal design.
D) To create the initial design metaphor for the project.

Source: IWkKAZ0jatM | Difficulty: Medium

Q18. In the IFPUG function point method, how is the Value Adjustment Factor (VAF) calculated?

A) By multiplying the number of user inputs by the number of user outputs.
B) By summing the weights of all functional units.
C) By using the formula VAF = 0.65 + (0.01 * TDI), where TDI is the Total Degree of Influence.
D) By dividing the estimated lines of code by the number of function points.

Source: MzhN6eZIUNU | Difficulty: Medium

Q19. What is a subcritical path and why is it important for project managers to monitor?

A) It is a path with negative float, indicating the project is already late.
B) It is the shortest path in the network, representing the fastest possible completion.
C) It is a path where activities have a very small, but non-zero, amount of float, which could easily become critical.
D) It is a path that has been intentionally delayed to save costs.

Source: lFu77P8s2QQ | Difficulty: Medium

Q20. What is a primary advantage of a project organization structure in terms of job satisfaction?

A) It allows specialists to focus on a single type of task, enhancing their expertise.
B) It provides job variety and rotation, as team members engage in diverse activities.
C) It guarantees lifetime employment on a single project.
D) It minimizes communication, allowing developers to work in isolation.

Source: MpNjo7bhBTg | Difficulty: Medium

Q21. What is the primary advantage of a Time and Material contract for the customer?

A) The total project cost is fixed and known upfront.
B) The supplier has a strong incentive to be cost-effective.
C) It is easy to change or add new requirements during development.

D) It shifts all financial risk to the supplier.
Source: olrW1uKQe0sM | Difficulty: Medium

Q22. In the context of algorithmic estimation, what is the purpose of 'model calibration'?

A) To select the programming language for the project.
B) To create a detailed Work Breakdown Structure (WBS).
C) To customize a generic model to fit specific organizational needs and project characteristics.
D) To calculate the final price charged to the customer.
Source: U5v_uKQe0Ic | Difficulty: Medium

Q23. In the contract tendering process, what is the primary purpose of establishing a Memoranda of Agreement (MoA)?

A) To finalize the payment schedule and contract price with the chosen vendor.
B) To invite as many suppliers as possible to bid on the project.
C) To freeze the technical specifications so that subsequent tenders can be judged primarily on price.
D) To legally bind all potential bidders to a non-disclosure agreement.
Source: -hlbuCXjNGY | Difficulty: Hard

Q24. Why is Weinberg's Zeroth Law of Reliability ('a software project that does not have to meet a reliability requirement can meet any other requirement') a critical principle in project estimation?

A) It implies that reliability is the easiest requirement to achieve.
B) It suggests that if reliability is not a strict requirement, other aspects might be prioritized at its expense, affecting true project cost and quality.
C) It proves that over-estimating project duration always leads to better reliability.
D) It states that reliability can only be estimated after the project is 100% complete.
Source: 0mpuhlabOWE | Difficulty: Hard

Q25. What is the defining characteristic of the Cleanroom technique for bug removal?

A) It relies exclusively on automated testing tools to find all defects.
B) It requires managers to personally review and approve every line of code.
C) Programmers are strictly not allowed to test their code by executing it for bug removal.
D) It is an informal process where developers mentally simulate code execution.
Source: 4ZLbpoBcdo8 | Difficulty: Hard

Q26. Why is Probability of Failure On Demand (POFOD) often a more relevant reliability metric for software systems than time-based metrics like MTTF?

A) POFOD is easier to calculate without any user data.
B) Software often operates on demand rather than continuously, making the success rate of individual invocations more meaningful.
C) POFOD accounts for the physical wear and tear of software components.
D) POFOD is the only metric that can be used for hardware reliability.
Source: 5iuEGNEyAPU | Difficulty: Hard

Q27. Why is the Spiral Model considered a 'meta-model'?

A) It is the oldest and most foundational of all lifecycle models.
B) It can only be applied to metaphysical or abstract software problems.
C) It is exclusively focused on risk handling and ignores other development activities.
D) It can incorporate aspects of waterfall, iterative, and prototyping models within its loops.
Source: sQyR78dlSzc | Difficulty: Hard

Q28. How is Risk Reduction Leverage (RRL) used to make decisions about which risk handling techniques to implement?

A) A higher RRL indicates a risk is more severe and should be avoided at all costs.
B) An RRL greater than 1 indicates the technique is cost-effective because the reduction in risk exposure outweighs the cost.
C) RRL is calculated by dividing the cost of the technique by the potential damage.
D) An RRL less than 1 means the risk handling technique is guaranteed to eliminate the risk.

Source: PehmL5jOmW8 | Difficulty: Hard

Q29. What is the key difference between a 'version' and a 'revision' in the context of Software Configuration Management (SCM)?

A) A version refers to source code, while a revision refers to documentation.
B) A version is a specific configuration of the entire product at a point in time, while a revision is a successive state of a single configuration item.
C) A version is created by the project manager, while a revision is created by a junior developer.
D) Versions are used for internal tracking, while revisions are released to customers.

Source: uITAo7fbm3M | Difficulty: Hard

Q30. According to Putnam's model, what is the expected impact on total effort if a project's development time is compressed by 50%?

A) The effort will decrease by 50%.
B) The effort will double (increase by 2 times).
C) The effort will quadruple (increase by 4 times).
D) The effort will increase by 16 times.

Source: _KdYrzzMuBo | Difficulty: Hard

Answer Key

1: B	2: C	3: B	4: B	5: C
6: B	7: B	8: C	9: B	10: C
11: C	12: C	13: B	14: D	15: C
16: B	17: C	18: C	19: C	20: B
21: C	22: C	23: C	24: B	25: C
26: B	27: D	28: B	29: B	30: D